

Term 3 — Lesson 1 Practice Activity (C6) with Solutions

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Evaluated criteria

- C6: Identifies and interprets Type I and Type II errors in a hypothesis test about a population proportion.

1 General Problem: decision outcomes and error ladder

A policy team reviews a hypothesis-test report about a population parameter (mean or proportion). The report provides either a direct decision or a p-value that must be read against one or more significance levels. Your goal is to interpret the decision in context without recalculating the test statistic.

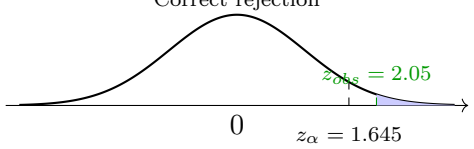
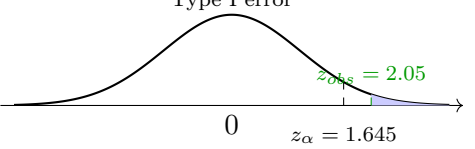
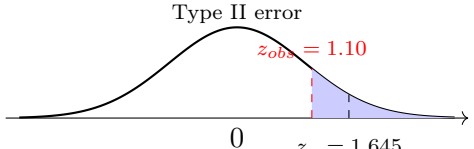
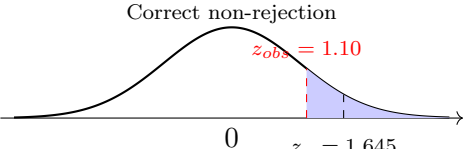
Use the following framework in every interpretation:

- If the report says *reject* H_0 , the immediate statistical risk is a **Type I error** (rejecting H_0 when H_0 is true).
- If the report says *fail to reject* H_0 , the immediate statistical risk is a **Type II error** (failing to reject H_0 when H_0 is false).
- Changing α changes decision strictness: smaller α gives stronger protection against Type I error but can increase exposure to Type II error.
- A statistical decision is evidence-based, not proof with certainty.

Decision-outcome table with explanations

	H_0 is false	H_0 is true
Reject H_0	Correct decision. The evidence leads to rejection, and in reality the null hypothesis is false. This is the desired outcome when the alternative is true.	Type I error. We reject H_0 even though H_0 is actually true. This is a false positive.
Fail to reject H_0	Type II error. We do not reject H_0 even though H_0 is actually false. This is a missed real effect.	Correct decision. We do not reject H_0 , and in reality the null hypothesis is true. This is the desired cautious outcome when there is no real effect.

Decision-outcome table with graph representations

	H_0 is false	H_0 is true
Reject H_0	Correct rejection  The observed statistic falls in the rejection region, and H_0 is false.	Type I error  The observed statistic falls in the rejection region, but H_0 is actually true.
Fail to reject H_0	Type II error  The observed statistic does not reach the rejection region, even though H_0 is false.	Correct non-rejection  The observed statistic does not reach the rejection region, and H_0 is true.

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- **Core interpretation rule:** decision first, then error mapping. Reject $H_0 \rightarrow$ immediate Type I concern; fail to reject $H_0 \rightarrow$ immediate Type II concern.
- **Certainty rule:** reject H_0 supports the alternative claim but does not prove it with certainty; fail to reject H_0 does not prove H_0 is true.
- **Threshold rule:** stricter α (for example 1%) reduces tolerance for false positives, while looser α (for example 10%) reduces tolerance for missed true effects.

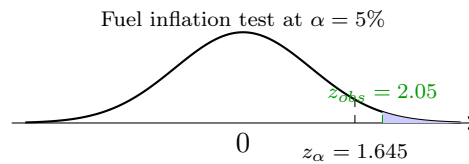
2 Fuel-price monitoring report (foundational reject case)

A city transport authority tests whether mean monthly fuel inflation is above the 4.0% value used in the fare budget. The report states

$$H_0: \mu = 4.0, \quad H_1: \mu > 4.0, \quad \alpha = 0.05,$$

and the reported decision is *reject* H_0 .

Interpret the decision for planners. State the supported claim, the immediate error risk, and what overclaim must be avoided.



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- **Reported decision:** reject H_0 .
- **Supported conclusion:** evidence supports that the population mean fuel inflation is above 4.0%.
- **Immediate error risk:** Type I error. In context, this means the authority could announce inflation is above 4.0% when the true mean is actually at the 4.0% benchmark.
- **What to avoid:** do not claim certainty and do not claim every fuel product has inflation above 4.0%; the inference is about the population mean.

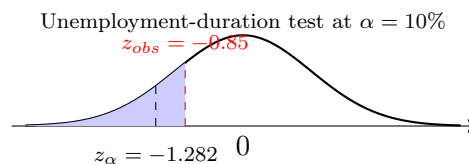
3 Public employment office review (foundational fail-to-reject case)

A labor ministry evaluates whether a job-matching reform reduced average unemployment duration below the 11-week benchmark:

$$H_0: \mu = 11, \quad H_1: \mu < 11, \quad \alpha = 0.10.$$

The report gives *fail to reject* H_0 .

Interpret this outcome and identify the immediate error concern.



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- **Reported decision:** fail to reject H_0 .
- **Supported conclusion:** evidence is insufficient to conclude that the population mean unemployment duration is below 11 weeks at the 10% level.
- **Immediate error risk:** Type II error. In context, a real reduction below 11 weeks may exist, but the ministry may fail to detect it.

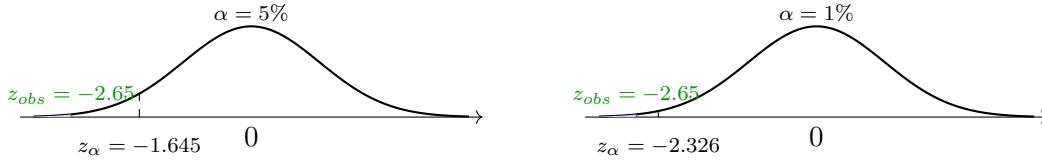
- **What to avoid:** do not say the reform had no effect or that $\mu = 11$ is proven true.

4 Export quality audit under strict and lenient standards

A manufacturer tests whether the true defect proportion is below the contractual reference 5%:

$$H_0: p = 0.05, \quad H_1: p < 0.05,$$

with reported p-value = 0.004. Interpret at $\alpha = 0.05$ and $\alpha = 0.01$.



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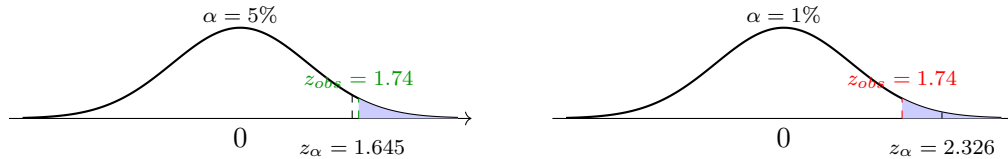
- **At $\alpha = 0.05$:** reject H_0 , so evidence supports $p < 0.05$.
- **At $\alpha = 0.01$:** reject H_0 , so evidence still supports $p < 0.05$ under the stricter threshold.
- **Immediate error concern (both decisions):** Type I error remains the relevant risk because the decision is reject H_0 at both thresholds.
- **Robustness interpretation:** the conclusion is robust across strict and lenient standards, but still not certainty.

5 Digital payment adoption with threshold-sensitive evidence

A central bank tests whether regular digital-payment use is above the 70% policy benchmark:

$$H_0: p = 0.70, \quad H_1: p > 0.70,$$

with reported p-value = 0.041. Compare $\alpha = 0.05$ and $\alpha = 0.01$.



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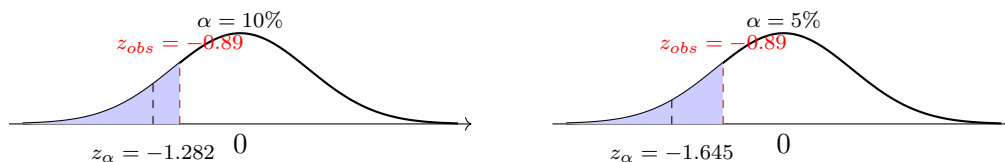
- **At $\alpha = 0.05$:** reject H_0 , so evidence supports $p > 0.70$. Immediate concern: Type I error.
- **At $\alpha = 0.01$:** fail to reject H_0 , so evidence is insufficient under a stricter rule. Immediate concern: Type II error.
- **Threshold lesson:** tightening α changed the decision and shifted the immediate practical concern from false-positive risk to missed-improvement risk.

6 Transport reliability monitoring with stable non-rejection

A rail agency checks whether average delay is below a 7.0-minute benchmark:

$$H_0: \mu = 7.0, \quad H_1: \mu < 7.0,$$

with p-value = 0.186. Interpret at $\alpha = 0.10$ and $\alpha = 0.05$.



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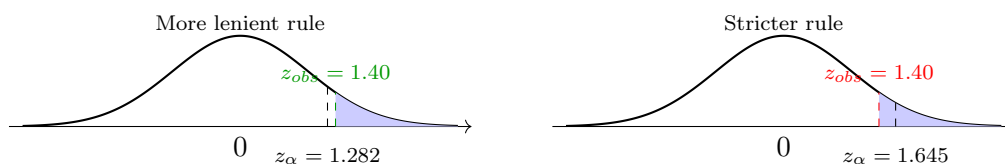
- **At $\alpha = 0.10$:** fail to reject H_0 .
- **At $\alpha = 0.05$:** fail to reject H_0 .
- **Immediate error concern (both decisions):** Type II error, because a true reduction in delay could exist but remain undetected.
- **Interpretation:** non-rejection is stable across thresholds, so evidence for improvement is weak under both standards.

7 Public health screening decision design

A ministry evaluates a low-cost screening program for early disease detection.

- The tested claim is that the detection rate is above the historical benchmark.
- A Type I error would trigger large emergency spending when the true detection rate has not improved.
- A Type II error would delay treatment scale-up when detection actually has improved.

Explain which error is more harmful in this context and justify whether a stricter or more lenient α is more appropriate.



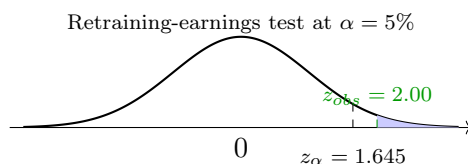
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- **Type I meaning:** false alarm of improvement, causing unnecessary emergency expansion costs.
- **Type II meaning:** missed real improvement, delaying access to beneficial treatment pathways.
- **Context-based judgment:** when missing real early detection is medically costly, planners may justify a somewhat more lenient α to reduce Type II risk, while still monitoring false positives through follow-up diagnostics.
- **Decision principle:** the choice of α reflects policy trade-offs, not a purely mechanical rule.

8 Labor policy evaluation and retraining expansion

A ministry considers expanding a retraining program. The test examines whether average post-training earnings exceed a benchmark. The report at $\alpha = 0.05$ says *reject H_0* .

Discuss the practical consequences of Type I and Type II errors and indicate which should receive greater caution for budget planning.



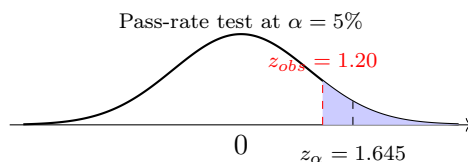
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- **Reported decision:** reject H_0 , so evidence supports higher mean earnings after retraining.
- **Immediate error concern:** Type I error. The ministry might scale up an ineffective program if the true gain is not above benchmark.
- **Type II consequence (counterfactual planning concern):** with a stricter threshold, the ministry could miss a genuinely effective program and underinvest in workers.
- **Policy emphasis:** if expansion is very costly and hard to reverse, stronger caution against Type I error is justified; if unemployment costs are severe, planners may tolerate slightly higher Type I risk to avoid Type II misses.

9 Educational intervention review with precision notes

An education board evaluates whether a reading-support intervention increased the pass rate above 68%. A sample of $n = 1600$ schools yields a margin of error of $\pm 2.4\%$, and the test report at $\alpha = 0.05$ says *fail to reject H_0* .

Interpret the decision, include the immediate error risk, and explain how sample size and margin of error affect confidence in interpretation.



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- **Reported decision:** fail to reject H_0 .
- **Supported conclusion:** evidence is insufficient to claim the true pass rate is above 68% at 5% significance.
- **Immediate error concern:** Type II error, meaning a true improvement above 68% may exist but was not detected.
- **Precision interpretation:** a large sample and relatively small margin of error make estimates more precise, but they still do not convert non-rejection into proof that no improvement exists.

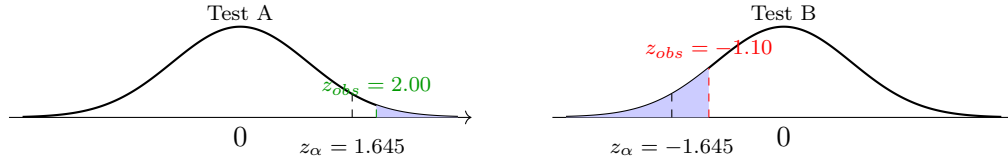
10 Service regulation dashboard with two independent tests

A regulator reports two independent service checks.

Test A (consumer complaints). $H_0: \mu = 24$ hours vs $H_1: \mu > 24$, $\alpha = 0.05$, decision: *reject H_0* .

Test B (network outage rate). $H_0: p = 0.030$ vs $H_1: p < 0.030$, $\alpha = 0.05$, decision: *fail to reject H_0* .

Interpret both tests separately and identify the immediate error concern for each.



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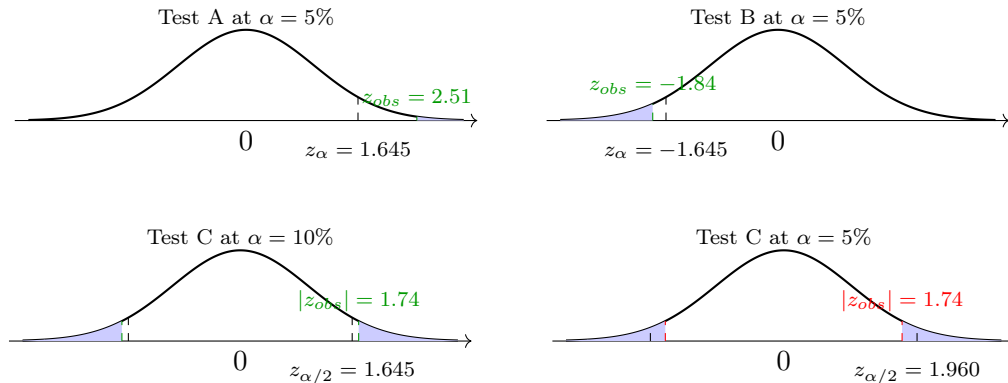
- **Test A decision:** reject H_0 . Evidence supports that mean complaint response time exceeds 24 hours.
- **Test A immediate error concern:** Type I error (false claim of slower response than benchmark).
- **Test B decision:** fail to reject H_0 . Evidence is insufficient to claim outage rate is below 3.0%.
- **Test B immediate error concern:** Type II error (true improvement in outage rate may be missed).
- **Dashboard lesson:** one report can contain different decisions and different error priorities across indicators.

11 Economic monitoring dashboard with three independent tests

A planning office reviews three indicators with p-values interpreted at $\alpha = 0.10$ and $\alpha = 0.05$.

- **Test A (fuel inflation above 4.0%):** p-value = 0.006.
- **Test B (unemployment duration below 11 weeks):** p-value = 0.033.
- **Test C (broadband access different from 76%):** p-value = 0.081.

Classify each finding as robust or threshold-sensitive and explain which error concern dominates for each decision.



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- **Test A:** reject at 10% and 5% $\rightarrow$ robust rejection; immediate concern Type I at both thresholds.

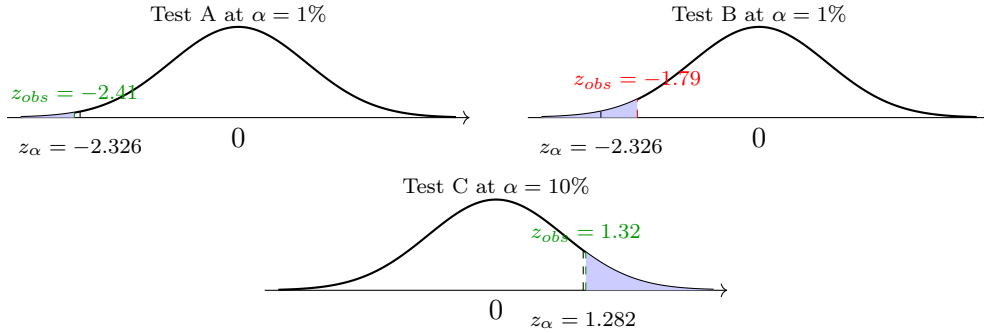
- **Test B:** reject at 10% and 5% → robust rejection; immediate concern Type I at both thresholds.
- **Test C:** reject at 10% but fail to reject at 5% → threshold-sensitive finding; Type I concern under 10% decision and Type II concern under 5% decision.
- **Communication implication:** robust findings can be reported with stronger confidence language; threshold-sensitive findings should be reported as conditional on confidence standard.

12 Consumer protection and quality assurance synthesis

A national report combines three independent tests, each reviewed at $\alpha = 0.10, 0.05$, and 0.01 :

- **Test A (unsafe-product complaint rate lower than benchmark):** p-value = 0.008.
- **Test B (factory defect rate lower than benchmark):** p-value = 0.037.
- **Test C (service-access equity improved above benchmark):** p-value = 0.094.

Classify each as robust, moderate, or tentative evidence, then connect each class to likely Type I/Type II consequence concerns.



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- **Test A (0.008):** reject at 10%, 5%, and 1% → **robust evidence**. Immediate concern remains Type I, but risk is controlled even under strict standard.
- **Test B (0.037):** reject at 10% and 5%, fail to reject at 1% → **moderate evidence**. Type I caution is still relevant, especially for high-stakes enforcement claims.
- **Test C (0.094):** reject only at 10% → **tentative evidence**. With stricter standards, Type II concern becomes central because possible improvement may remain unconfirmed.
- **Synthesis:** evidence class should guide communication strength and policy reversibility.

13 Integrated policy planning under alternative error tolerances

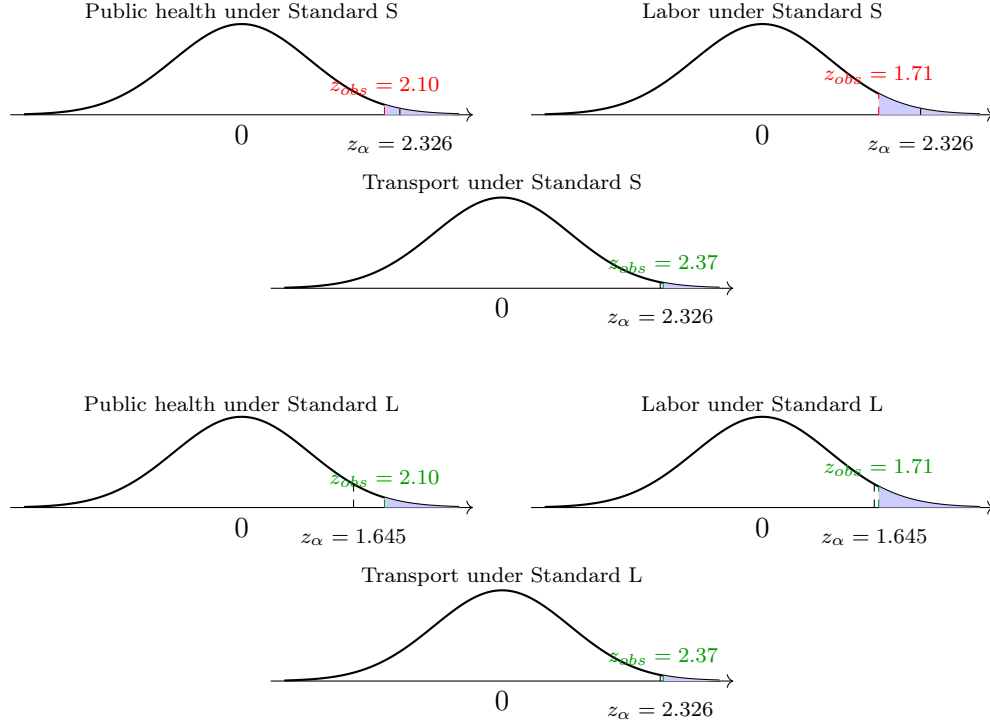
A cabinet committee must choose one of two reporting standards for next year's action plan:

- **Standard S (stricter):** use $\alpha = 0.01$ for all flagship claims.
- **Standard L (less strict):** use $\alpha = 0.05$ for program-expansion claims and $\alpha = 0.01$ only for high-cost irreversible actions.

Current p-values are:

- Public-health early detection improvement: $p = 0.018$.
- Labor retraining earnings improvement: $p = 0.044$.
- Safety-critical transport failure increase: $p = 0.009$.

For each standard, state which claims are supported. Then justify which standard better balances Type I and Type II costs across these domains.



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- **Under Standard S ($\alpha = 0.01$ everywhere):**
 - Public health (0.018): fail to reject H_0 .
 - Labor retraining (0.044): fail to reject H_0 .
 - Transport safety risk (0.009): reject H_0 .

Immediate concern is mostly Type II for the first two domains and Type I for the transport-safety rejection.

- **Under Standard L (mixed rule):**
 - Public health at 5%: reject H_0 .
 - Labor retraining at 5%: reject H_0 .
 - Transport safety at 1%: reject H_0 .

This rule accepts more Type I exposure for expansion claims to reduce Type II misses, while keeping strict Type I protection for irreversible safety actions.

- **Policy judgment:** Standard L is often preferable when missing true beneficial programs is costly, but Standard S is preferable when false positives are extremely costly or politically irreversible.